# BOQ

| HVAC WORKS | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | ACTUAL QTY REQUIRED FOR MODIFICATION WORK FOR TRANCH-1 | | | | | Dismantle work BOQ For Tranch-1 | | | | REUSE MATERIAL | SCRAP MATERIAL | Material in Store (B) | Actual qty required for Tranch-1 additional work | | |
| Sr.No | Description | Unit | 1 F | Outside Shaft for 1F | Terrace | Shaft 1/2/3/4 | Total Qty | 1 F | V S | Terrace | Total Qty | QTY | QTY | QTY | SUPPLY | Installation | Remarks |
| 1 | **BUFFER TANK** | | | | | | 0 | | | | | | 0 | | | | |
| 1.1 | Supply, installation, testing and commissioning of **Horizontal type chilled water buffer tank** complete with carbon steel, Baffles. designed in accordance with ASME Section VIII Div 1 ED 2019 Standard (ASME stamping not required). Insulation and Aluminum Cladding 26 gauge including all accessories and shall be suitable for outdoor installation. The tank must be constructed in accordance with section VIII of the ASME Boiler and pressure vessel (Unfired) code. Tanks having usable storage capacity as per requirement mentioned below. (Tank manufacturer shall consider the Figure of Merit (FOM), Number of Baffle Plates and Baffle arrangement shall be suitable to meet the required backup time). Maximum Working Pressure to be considered as 6 Kg/Cm2 where the Tanks will be installed on True Terrace Platform at outlet of Primary Variable Pump discharge side of piping distribution (please refer P&ID).<br>The tank shall be insulated with Nitrile rubber Insulation (Thickness selection should be submitted) Insulation material & thickness to match the design requirement & temperature. Buffer tank should be provided with Baffles / diffusers to achieve required thermal backup time. The joints of aluminium sheet shall be overlapped on both longitudinal and transverse direction and shall be riveted. Buffer tank to be sized considering water storage temperature of 18.0 Deg C where as the return Chilled Water will be at 26.0 deg.C with Ambient of 42.7 Deg C (ASHRAE N=50 for Mumbai). Tank shall have Joint Efficiency=1.0 with 100%. Radiographic Test, surface preparation & painting for desired finish and quality assurance shall be included.<br>Vendor to provide weight of empty tank, and completely filled tank weight. Vendor to provide the thickness calculation sheet along with technical submittal. Vendor to submit the CFD analysis report for Thermal Storage Tank, ensuring required backup is achieved. | | | | | | 0 | | | | | | 0 | | | | |
| | Provide the below provision along with Tank:<br>Detachable Ladder & Platform arrangement above the Tank suitable be installed at site. Height to be mentioned in drawing.<br>Manhole- 600 x 600mm (No. of Manholes depends on the size of tank as per manufacturers recommendation)<br>Drain: 50 mm dia.<br>Air purge valve with ball valve isolation<br>Safety release valve with ball valve isolation<br>Require number of thermowells. Thermowell provision should ensure to install Temperature sensor at least 300 mm inside the tank without any obstruction. Based on the Insulation Thickness and Tapping length outside the Tank surface, the Sensor length will be decided. One Sensor to measure inlet water temperature and one sensor to measure outlet water temperature of the tank. rest of the sensors as per baffle arrangement, to measure intermediate temperature. Number of sensor points shall be as per OEM to provide sufficient detail regarding % charge level of tank in steps of 10% | | | | | | | | | | | | | | | | |
| 1.2 | Tank shall be as per below specifications:<br>**Buffer Tank dia. Should be max 2500 mm.**<br>**Buffer Tank Length should be max 7000 mm.**<br>C/C distance for the saddles should be restricted to 3300mm<br>**Inlet: 250mm dia**<br>**Outlet: 250mm dia**<br>Drain: 50 mm dia.<br>Test pressure: 8 kg/sq.cm<br>**Volume: 30 KL**<br>**Backup time: 5 minutes i.e. 18 Deg C at outlet at a flow of 875 USGPM for 5 minutes**<br>**TES Tank shall be compatible for supporting from Super Terrace Platform since supporting on existing Terrace Floor is not possible** | Nos | | | 1 | | 1 | | | | 0 | | 0 | | 1 | 1 | |

| HVAC WORKS | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | ACTUAL QTY REQUIRED FOR MDIFICATION | | Dismantle work BOQ For Tranch-1 | | REUSE MATERIAL | SCRAP MATERIAL | Material in Store (B) | Actual qty required for Tranch-1 additional | | TRANCHE 2 | | | |
| Sr.No | Description | Unit | 2 F | Total Qty | 2 F | Total Qty | QTY | QTY | QTY | SUPPLY | Installation | 4 F | Terrace | TOTAL QTY | Remarks |
| 1 | **BUFFER TANK** | | | 0 | | | | 0 | | | | | | | |
| 1.1 | Supply, installation, testing and commissioning of **Horizontal type chilled water buffer tank** complete with carbon steel, Baffles. designed in accordance with ASME Section VIII Div 1 ED 2019 Standard (ASME stamping not required). Insulation and Aluminum Cladding 26 gauge including all accessories and shall be suitable for outdoor installation. The tank must be constructed in accordance with section VIII of the ASME Boiler and pressure vessel (Unfired) code. Tanks having usable storage capacity as per requirement mentioned below. (Tank manufacturer shall consider the Figure of Merit (FOM), Number of Baffle Plates and Baffle arrangement shall be suitable to meet the required backup time). Maximum Working Pressure to be considered as 6 Kg/Cm2 where the Tanks will be installed on True Terrace Platform at outlet of Primary Variable Pump discharge side of piping distribution (please refer P&ID).<br>The tank shall be insulated with Nitrile rubber Insulation (Thickness selection should be submitted) Insulation material & thickness to match the design requirement & temperature. Buffer tank should be provided with Baffles / diffusers to achieve required thermal backup time. The joints of aluminium sheet shall be overlapped on both longitudinal and transverse direction and shall be riveted. Buffer tank to be sized considering water storage temperature of 22.0 Deg C where as the return Chilled Water will be at 30.6 deg.C with Ambient of 42.7 Deg C (ASHRAE N=50 for Mumbai). Tank shall have Joint Efficiency=1.0 with 100%. Radiographic Test, surface preparation & painting for desired finish and quality assurance shall be included.<br>Vendor to provide weight of empty tank, and completely filled tank weight. Vendor to provide the thickness calculation sheet along with technical submittal. Vendor to submit the CFD analysis report for Thermal Storage Tank, ensuring required backup is achieved. | | | 0 | | | | 0 | | | | | | | |
| | Provide the below provision along with Tank:<br>Detachable Ladder & Platform arrangement above the Tank suitable be installed at site. Height to be mentioned in drawing.<br>Manhole- 600 x 600mm (No. of Manholes depends on the size of tank as per manufacturers recommendation)<br>Drain: 50 mm dia.<br>Air purge valve with ball valve isolation<br>Safety release valve with ball valve isolation<br>Require number of thermowells. Thermowell provision should ensure to install Temperature sensor at least 300 mm inside the tank without any obstruction. Based on the Insulation Thickness and Tapping length outside the Tank surface, the Sensor length will be decided. One Sensor to measure inlet water temperature and one sensor to measure outlet water temperature of the tank. rest of the sensors as per baffle arrangement, to measure intermediate temperature. Number of sensor points shall be as per OEM to provide sufficient detail regarding % charge level of tank in steps of 10% | | | | | | | | | | | | | | |
| 1.2 | Tank shall be as per below specifications:<br>**Buffer Tank dia. Should be max 2500 mm.**<br>**Buffer Tank Length should be max 7000 mm.**<br>C/C distance for the saddles should be restricted to 3300mm<br>**Inlet: 250mm dia**<br>**Outlet: 250mm dia**<br>Drain: 50 mm dia.<br>Test pressure: 8 kg/sq.cm<br>**Volume: 40 KL**<br>**Backup time: 5 minutes i.e. 22.0 Deg C at outlet at a flow of 875 USGPM for 5 minutes** | Nos | | | | | | 0 | | - | - | | | - | |

| HVAC WORKS |
|---|

| Sr.No | Description | Unit | ACTUAL QTY REQUIRED FOR MDIFICATION WORK FOR | | Dismantle work BOQ For Tranch-1 | | REUSE MATERIAL | SCRAP MATERIAL | Material in Store (B) | Actual qty required for Tranch-1 additional work | | TRANCHE 2 | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 3 F | Total Qty | 3 F | Total Qty | QTY | QTY | QTY | SUPPLY | Installation | 5 F | TOTAL QTY | |
| 1 | **BUFFER TANK** | | | 0 | | | | - | | | | | | |
| 1.1 | Supply, installation, testing and commissioning of **Horizontal type chilled water buffer tank** complete with carbon steel, Baffles. designed in accordance with ASME Section VIII Div 1 ED 2019 Standard (ASME stamping not required). Insulation and Aluminum Cladding 26 gauge including all accessories and shall be suitable for outdoor installation. The tank must be constructed in accordance with section VIII of the ASME Boiler and pressure vessel (Unfired) code. Tanks having usable storage capacity as per requirement mentioned below. (Tank manufacturer shall consider the Figure of Merit (FOM), Number of Baffle Plates and Baffle arrangement shall be suitable to meet the required backup time). Maximum Working Pressure to be considered as 6 Kg/Cm2 where the Tanks will be installed on True Terrace Platform at outlet of Primary Variable Pump discharge side of piping distribution (please refer P&ID).<br>The tank shall be insulated with Nitrile rubber Insulation (Thickness selection should be submitted) Insulation material & thickness to match the design requirement & temperature. Buffer tank should be provided with Baffles / diffusers to achieve required thermal backup time. The joints of aluminium sheet shall be overlapped on both longitudinal and transverse direction and shall be riveted. Buffer tank to be sized considering water storage temperature of 22.0 Deg C where as the return Chilled Water will be at 30.6 deg.C with Ambient of 42.7 Deg C (ASHRAE N=50 for Mumbai). Tank shall have Joint Efficiency=1.0 with 100%. Radiographic Test, surface preparation & painting for desired finish and quality assurance shall be included.<br>Vendor to provide weight of empty tank, and completely filled tank weight. Vendor to provide the thickness calculation sheet along with technical submittal. Vendor to submit the CFD analysis report for Thermal Storage Tank, ensuring required backup is achieved. | | | 0 | | | | - | | | | | | |
| | Provide the below provision along with Tank:<br>Detachable Ladder & Platform arrangement above the Tank suitable be installed at site. Height to be mentioned in drawing.<br>Manhole- 600 x 600mm (No. of Manholes depends on the size of tank as per manufacturers recommendation)<br>Drain: 50 mm dia.<br>Air purge valve with ball valve isolation<br>Safety release valve with ball valve isolation<br>Require number of thermowells. Thermowell provision should ensure to install Temperature sensor at least 300 mm inside the tank without any obstruction. Based on the Insulation Thickness and Tapping length outside the Tank surface, the Sensor length will be decided. One Sensor to measure inlet water temperature and one sensor to measure outlet water temperature of the tank. rest of the sensors as per baffle arrangement, to measure intermediate temperature. Number of sensor points shall be as per OEM to provide sufficient detail regarding % charge level of tank in steps of 10% | | | | | | | | | | | | | |
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**HVAC WORKS**

| Sr.No | Description | Unit | ACTUAL QTY REQUIRED FOR MDIFICATION | | Dismantle work BOQ For Tranch-1 | | REUSE MATERIAL | SCRAP MATERIAL | Material in Store (B) | Actual qty required for Tranch-1 additional | | TRANCHE 2 | | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 7 F | Total Qty | 7 F | Total Qty | QTY | QTY | QTY | SUPPLY | Installation | 6 F | TOTAL QTY | |
| 1 | **BUFFER TANK** | | | 0 | | | | 0 | | | | | | |
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| | Provide the below provision along with Tank:<br>Detachable Ladder & Platform arrangement above the Tank suitable be installed at site. Height to be mentioned in drawing.<br>Manhole- 600 x 600mm (No. of Manholes depends on the size of tank as per manufacturers recommendation)<br>Drain: 50 mm dia.<br>Air purge valve with ball valve isolation<br>Safety release valve with ball valve isolation<br>Require number of thermowells. Thermowell provision should ensure to install Temperature sensor at least 300 mm inside the tank without any obstruction. Based on the Insulation Thickness and Tapping length outside the Tank surface, the Sensor length will be decided. One Sensor to measure inlet water temperature and one sensor to measure outlet water temperature of the tank. rest of the sensors as per baffle arrangement, to measure intermediate temperature. Number of sensor points shall be as per OEM to provide sufficient detail regarding % charge level of tank in steps of 10% | | | | | | | | | | | | | |
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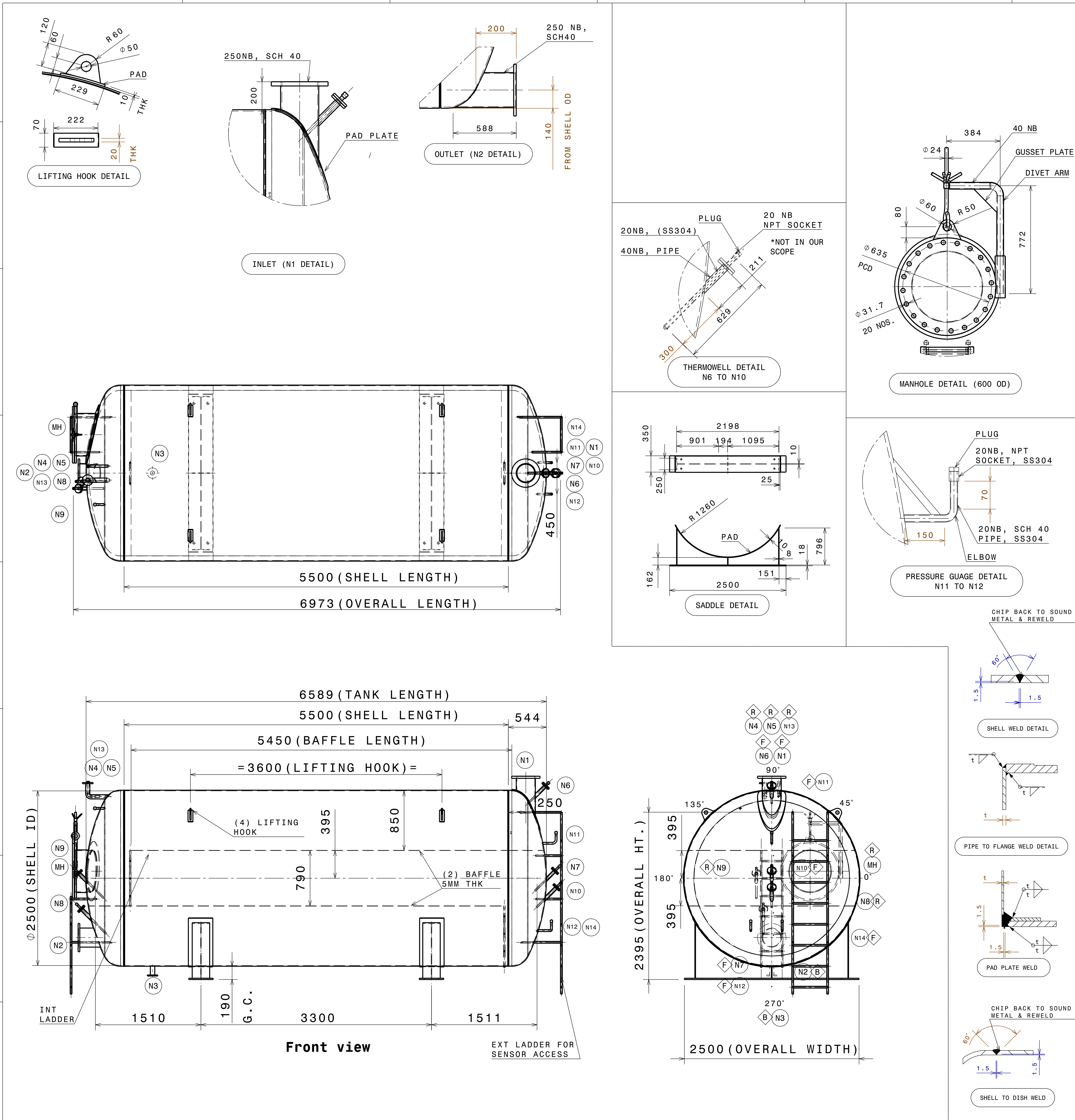
# MAKE LIST

| HVAC Components | | | |
|---|---|---|---|
| **Sr. No.** | **Equipment /Material** | **Approved Makes** | **Proposed Make** |
| 5 | Buffer Tank | Anergy / Sevcon / Ensaviour / NES | NES |

CATALOGUE / TECHNICAL SPECIFICATION




NOTES:

| | |
|---|---|
| 1 | ALL DIMENSIONS ARE IN MM UNLESS OTHERWISE SPECIFIED. |
| 2 | PAINTING OUTSIDE MS PARTS- TWO COAT OF RED OXIDE |
| 3 | PAINTING INSIDE- TWO COAT OF COAL TAR EPOXY |
| 4 | PROVIDE PAD PLATE FOR NOZZLES ABOVE 50 NB & RIB FOR BELOW 50 NB. |
| 5 | INSULATION CLADDING NOT IN SCOPE. |
| 6 | CLEAN THE SURFACES PROPERLY BEFORE WELDING. |

DESIGN CODE: ASME SECT. VIII, DIV I

| SR. NO. | ELEMENT | UNIT | DESCRIPTION |
|---|---|---|---|
| 1 | SPECIFIC GRAVITY OF LIQUID | GM/CC | 1 |
| 2 | WORKING TEMPERATURE | °C | 18 |
| 3 | OPERATING TYPE | | CONTINOUS |
| 4 | JOINT EFFICIENCY | % | 1 |
| 5 | RADIOGRAPHY | % | 100 |
| 6 | DESIGN PRESSURE | KG/CM2 | 6.5 |
| 7 | MAX. WORKING PRESSURE | KG/M2 | 6 |
| 8 | HYDRO PRESSURE | KG/CM2 | 8 |
| 9 | HYDRO CAPACITY | KL | 30.4 |
| 10 | WORKING VOLUME | KL | 30 |
| 11 | EMPTY WEIGHT | TON | 7 |
| 12 | WATER FILLED WEIGHT | TON | 37.4 |

NOZZLES SCHEDULE

| SR. NO. | NB | SCH. | PROJECTION | MOC PIPE | CLASS | SERVICE FOR |
|---|---|---|---|---|---|---|
| MH | 600 OD | - | 250 | SA106 | ASA #150 | MANHOLE |
| N1 | 250 | 40 | 200 | SA106 | ASA #150 | INLET |
| N2 | 250 | 40 | 200 | SA106 | ASA #150 | OUTLET |
| N3 | 50 | 40 | 150 | SA106 | ASA #150 | DRAIN |
| N4 | 25 | 40 | 150 | SA106 | ASA #150 | MAN. AIR VENT |
| N5 | 25 | 40 | 150 | SA106 | NPT HALF COUPLING | AUTO AIR VENT |
| N6, N7, N8, N9, N10 | 40 X 20 | 40 | 150 | SS304 | ASA #150 + NPT HALF COUPLING | THERMOWELL |
| N11, N12 | 20 | 40 | 200 | SS304 | BEND +NPT HALF COUPLING | PRESSURE GUAGE |
| N13, N14 | 20 | 40 | 200 | SS304 | BEND+NPT HALF COUPLING | PRESSURE TRANSMITTER |

| SR. NO. | PART | DESCRIPTION | QTY | MAT. |
|---|---|---|---|---|
| 12 | BASE PLATE | 16 MM | | MS |
| 11 | ALL NON WETTED PARTS | - | - | MS |
| 10 | PAD PLATE | 10 MM | - | SA516, GR B |
| 9 | NOZZLE PIPE | - | - | SA106/ SS304 |
| 8 | NOZZLE FLANGES | ASA#150 | - | IS2062 |
| 7 | INT. LADDER | 25 OD X 2MM THK | 1 | SS304 |
| 6 | EXT. LADDER | 25 NB X CLASS B | 1 | MS |
| 5 | LIFTING HOOK | 20 MM | 4 | IS2062 GR. B |
| 4 | SADDLES | - | 2 | IS2062, GR B |
| 3 | BAFFLES | 5 MM | 2 | IS2062, GR B |
| 2 | BOTH DISH | 10 MM | 2 | SA 516, GR B |
| 1 | SHELL | 10 mm | 1 | SA 516, GR B |

**BILL OF MATERIAL**



# NES INDIA ENGINEERS

| | | | | | |
|---|---|---|---|---|---|
| DRAWN BY: KS | DATE: 25/07/2024 | DRAWING TITLE: 30KL HORIZONTAL STORAGE TANK | | | |
| CHECKED BY: NAD | DATE | SIZE: A3 | DRAWING NUMBER: NIE- 593 | | REV: 03 |
| DESIGNED BY: NAD | DATE | SCALE: 1:1 | WEIGHT(kg): XXX | | SHEET: 1/1 |

**SUBJECT:: Technical Data Sheet FOR 30KL *CAPACITY THERMAL STORAGE TANK***

**Ssupply of Thermal water storage tanks having total capacity of 30 KL**

**Make : NES INDIA**

**Total Qty : 1 nos.**

## 1. TANK SPECIFICATIONS

| | |
|---|---|
| **TYPE OF TANK** | **HORIZONTAL CYLINDRICAL DISH ON BOTH SIDE** |
| | |
| **DESIGN BASIS** | **ASME SEC VIII DIV1. 2019** |
| SP GRAVITY OF LIQUID | 1 GM/CC |
| JOINT EFFICIENCY | 1 |
| TEMPERATURE | INLET 26 DEG CELCIUS OUTLET 18 DEG CELCIUS |
| OPERATION TYPE | CONTINUOUS |
| WORKING VOLUME | 30 KL |
| GROSS VOLUME CAPACITY | 30.4KL |
| DESIGN PRESSURE | 6.5 Kg/cm2 |
| WORKING PRESSURE | 6.0 kg/m2 |
| HYDRO TEST PRESSURE | 8.0 Kg/cm2 |
| RADIOGRAPHY | 100% CONSIDERED |
| | |
| **MATERIAL OF CONSTUCTION** | |
| SHELL | CS (SA516 GR70) |
| BOTH SIDE DISH | CS (SA516 GR70) |
| BAFFLE PLATE | IS2062 GR B |
| NOZZLES PIPES | SA106 |
| NOZZLES FLANGES | IS2062 |
| SADDLE SUPPORTS | MS (IS2062 GR B) |
| INTERNAL LADDER | SS304 |
| EXTERNAL LADDER & RAILING | MS |
| FASTNERS | MS PLATED |
| ALL WETTED PARTS | SA516 GR70 |
| ALL NON WETTED PARTS | MS |
| **TECHNICAL SPECIFICATIONS** | |
| SHELL DIA | 2500 MM |
| SHELL LENGTH (WL-WL) | 5500 MM |
| TOTAL LENGHTH- DISH TO DISH | 6570MM |
| SHELL THICKNESS | **10 MM** |
| BOTH DISH THICKNESS | **10 MM** |

| | |
|---|---|
| HORIZONTAL BAFFLE PLATE | 5 MM (2 nos equally spaced across the length to separate tank volume in Three parts |
| VERTICAL BAFFLE | 5MM THK, 100 width x 3 Nos in each section on all side |
| SUPPORT | 2 NOS SADDLE SUPPORT WITH BASE PLATE |
| CLOSER TYPE | 10% TORISPERICAL DISH ON BOTH ENDS |
| NOZZLES & MANHOLE | **1) 600 NB X 1 Nozzle with blind flange for manhole at side dish end as per client requirement**<br>**2) 250 NB x 2 nozzles- one for inlet and one for outlet**<br>**3) 50 NB x 1 nozzles for Mannual Drain.**<br>**4) 50 NB x 1 nozzle for manual air vent**<br>**5) 25 NB x 1Half couplings (NPT socket) nozzle for automatic air vent**<br>**6) 20NB x 4 nos. Thermowell for Temp sensor One at inlet, one at outlet and 2 at each section of partition**<br>**7) 20 NB X 2 nos. Socket provided for pressure gauge one at inlet and one at outlet side**<br>**8) 20 NB x 2nos for Pressure Transmitter one at inlet and one at outlet side** |
| OVERALL DIAMENSIONS | Dia-3000 X H-6900 mm APPROX |
| | |
| **SURFACE FINISH** | |
| OUTSIDE | TWO COAT OF RED OXIDE |
| INSIDE | TWO COAT OF COAL TAR EPOXY |
| **INSULATION & CLADDING** | 64MM NITRILE RUBBER WITH 25 SWG AL CLADDING |

**PRESSURE DROP CLACULATION:**

**INLET PIPE SECTION- Tank Inlet Nozzle**

**Calculation output:**

| | |
|---|---|
| Flow medium: | Water 20 °C / liquid |
| Volume flow: | 875 petr.bl./day |
| Weight density: | 998.206 kg/m3 |
| Dynamic Viscosity: | 1001.61 10-6 kg/ms |
| Element of pipe: | Sudden enlargement |
| Dimensions of element: | Diameter of pipe D1: 250 mm |
| | Diameter of pipe D2: 1440 mm |
| Velocity of flow: | 0.03 m/s |
| Reynolds number: | 8172 |
| Velocity of flow 2: | 0 m/s |
| Reynolds number 2: | 1419 |
| Flow: | turbulent |
| Absolute roughness: | |
| Pipe friction number: | |
| Resistance coefficient: | 1035.4 |
| Resist.coeff.branching pipe: | - |
| Press.drop branch.pipe: | - |
| Pressure drop: | 0.01 mbar |
| | 0 bar |

**Pressure drop at Inlet section is 0.01 mbar**

**OUTLET PIPE SECTION- Tank Outlet Pipe**

**Calculation output**

| | |
|---|---|
| Flow medium: | Water 20 °C / liquid |
| Volume flow: | 875 petr.bl./day |
| Weight density: | 998.206 kg/m3 |
| Dynamic Viscosity: | 1001.61 10-6 kg/ms |
| Element of pipe: | Sudden contraction |
| Dimensions of element: | Diameter of pipe D1: 1440 mm |
| | Diameter of pipe D2: 250 mm |
| Velocity of flow: | 0 m/s |
| Reynolds number: | 1419 |
| Velocity of flow 2: | 0.03 m/s |
| Reynolds number 2: | 8172 |
| Flow: | laminar |
| Absolute roughness: | |
| Pipe friction number: | |
| Resistance coefficient: | 0.59 |
| Resist.coeff.branching pipe: | - |
| Press.drop branch.pipe: | - |
| Pressure drop: | 0 mbar |
| | 0 bar |

**Pressure drop at Outlet section is 0 mbar**

## TANK SECTION:
## Calculation output

| | |
|---|---|
| Flow medium: | Water 20 °C / liquid |
| Volume flow: | 875 gal/min |
| Weight density: | 998.206 kg/m3 |
| Dynamic Viscosity: | 1001.61 10-6 kg/ms |
| Element of pipe: | circular |
| Dimensions of element: | Diameter of pipe D: 1440 mm |
| | Length of pipe L: 15 m |
| Velocity of flow: | 0.03 m/s |
| Reynolds number: | 48645 |
| Velocity of flow 2: | - |
| Reynolds number 2: | - |
| Flow: | turbulent |
| Absolute roughness: | 0.1 mm |
| Pipe friction number: | 0.02 |
| Resistance coefficient: | 0.22 |
| Resist.coeff.branching pipe: | - |
| Press.drop branch.pipe: | - |
| Pressure drop: | 0 mbar |
| | 0 bar |

**Total pressure drop across Tank= (0.01+0.00+00) = 0.01 Bar**

## VOLUME CALCULATION SHEET: -

| | | |
|---|---|---|
| tank dia ID | 2500 | mm |
| shell ht WL-WL | 5500 | mm |
| SF of Dish | 50 | mm |
| Grond clearnace | 700 | mm |
| dish ht with SF | 535 | mm |
| dish volume | 1732.608544 | Lits |
| shell volume | 26984.375 | Lits |
| **total volume** | 30449.59209 | Lits |
| total height of tank | 6570 | MM |

## SHELL THICKNESS CALCULATION:

| Design of Shell under Internal Pressure | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |
| **Material** | **MATL** | SA 516 Gr.70 | | | |
| **Design Pressure** | **P** | 92.625 | psi | 6.5 | kg / cm² |
| **Design Temperature** | **Temp** | 131 | ºF | 55 | ºC |
| | | | | | |
| **Corrosion Allowance** | **CA** | 0.06 | inch | 1.5 | mm |
| **Polishing Allowance** | **PA** | 0.00 | inch | 0 | mm |
| **Joint Efficiency** | **E** | 0.85 | | | |
| **Radiography** | | Spot + Tee | | | |
| **Shell I/D** | **I/D** | 98.43 | inch | 2500 | mm |
| **Provided Thickness** | **t** | 0.39 | inch | 10 | mm |
| **Stress at Design temp** | **S** | 20000 | psi | 1406.14 | kg / cm² |
| | | | | | |
| **Calc Thickness** | **tc** | 0.27 | inch | 6.83 | |
| **Minimum Thickness (tcal + CA + PA)** | **tm** | 0.33 | inch | 8.33 | |
| **Max All Working Press (Corroded Condition)** | **MAWP** | 115.13 | psi | 8.09 | kg / cm² |

**SELECTED SHELL THICKENSS- 10MM**

## DISH THICKENESS CALCUALTION:

| Design of Dish under Internal Pressure | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |
| **Material** | **MATL** | SA 516 Gr.70 | | | |
| **Design Pressure** | **P** | 92.625 | psi | 6.5 | kg / cm² |
| **Design Temperature** | **Temp** | 131 | ºF | 55 | ºC |
| | | | | | |
| **Joint Efficiency** | **E** | 0.85 | | | |
| **Corrosion Allowance** | **CA** | 0.06 | inch | 1.5 | mm |
| **Polishing Allowance** | **PA** | 0.00 | inch | 0 | mm |
| **Thinning Allowance** | **TA** | 0.06 | inch | 1.5 | mm |
| **Radiography** | | Spot + Tee | | | |
| **Dish Type** | | 2: 1 Semi Ellipsoidal | | | |
| **Dish I/D** | **I/D** | 98.43 | inch | 2500 | mm |
| **Crown Radius** | | Not Required | | 2500 | mm |
| **Knuckle Radius** | | Not Required | | 250 | mm |
| **Provided Thickness** | **t** | 0.39 | inch | 10 | mm |
| **Stress at Design temp** | **S** | 20000 | psi | 1406.14 | kg / cm² |
| | | | | | |
| **Calc Thickness** | **tc** | 0.27 | inch | 6.81437 | |
| **Minimum Thickness (tcal + CA + PA + TA)** | **tm** | 0.39 | inch | 9.81437 | |
| **Max All Working Pressure (Corroded Condition)** | **MAWP** | 115.52 | psi | 8.12 | kg / cm² |

**Selected dish end thickness- 10mm**